DATA SCIENCE

Task Presentation

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Data Science

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ACKNOWLEDGEMENT

I would like to express my gratitude to ShadowFox for giving me the opportunity to present the tasks I have completed during this internship.

I would also like to extend my thanks to my coordinator, Mr. Aakash, for guiding me throughout this experience. I acknowledge that the tasks were completed independently by me. Additionally, I utilized certain reference materials for research purposes and used ChatGPT for generating reference ideas.

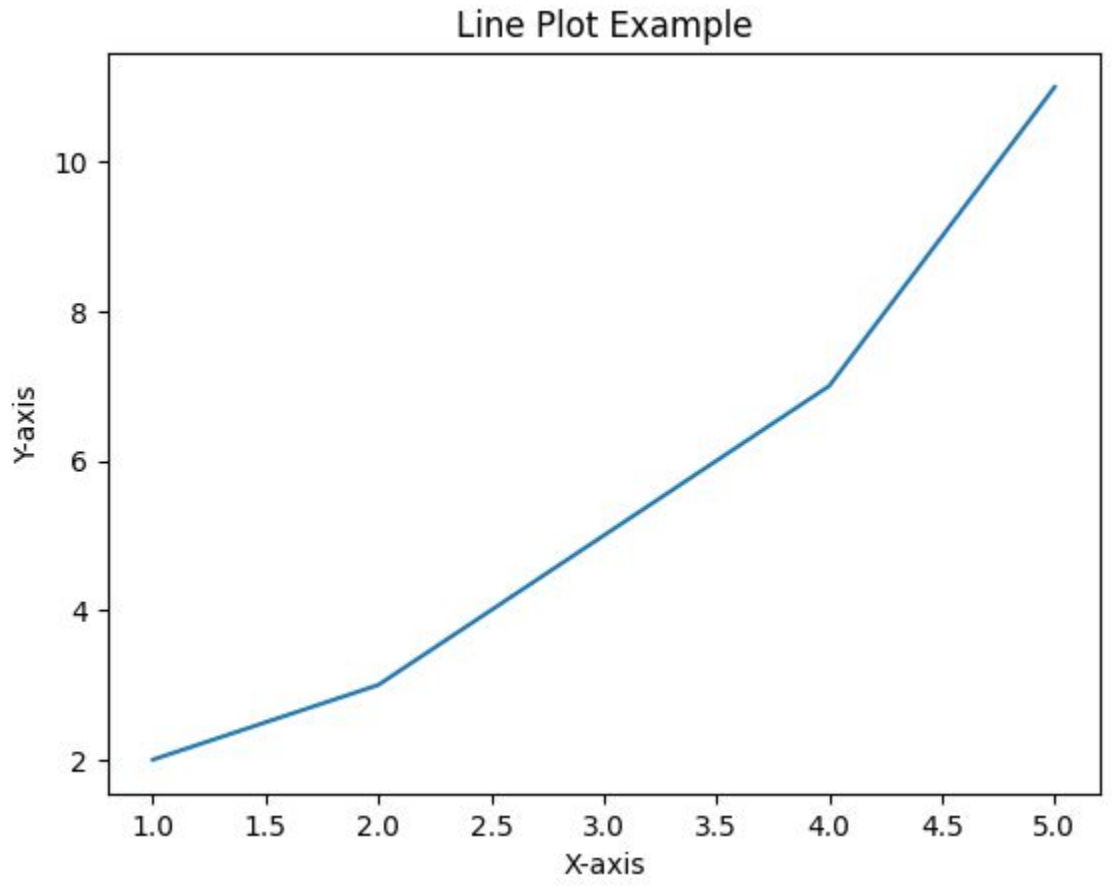
BEGINNER LEVEL

Task: Comprehensive Analysis of Data Visualization Techniques using Matplotlib and Seaborn

- Analyzed data visualization techniques using Matplotlib and Seaborn in Python.
- Created various types of visualizations
  - line plot
  - Bar chart
  - Scatter plot
  - Histogram
  - Pie chart
  - Box plot
  - Pair plot

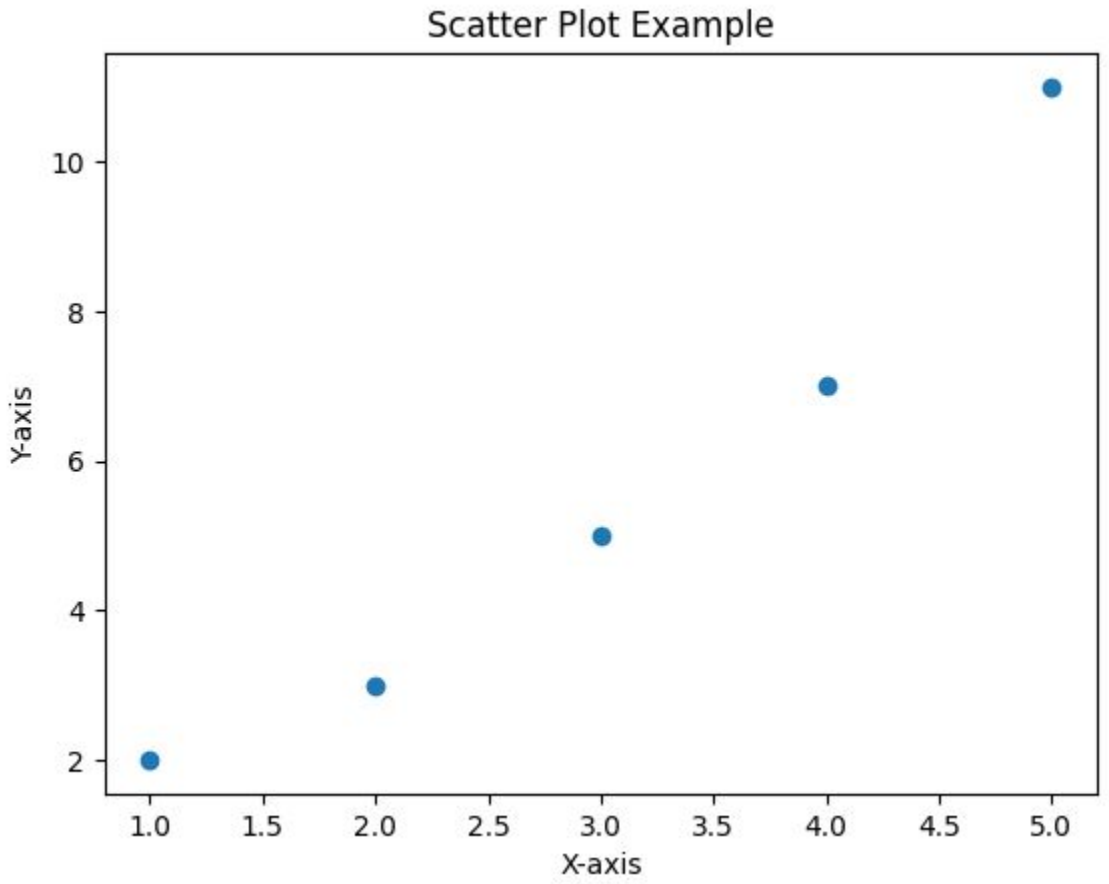
Line Plot

- **Description:** Line plots are used to represent data points connected by straight lines. They are useful for visualizing trends over time.
- **Use Case:** Analyzing time-series data or trends over continuous intervals.



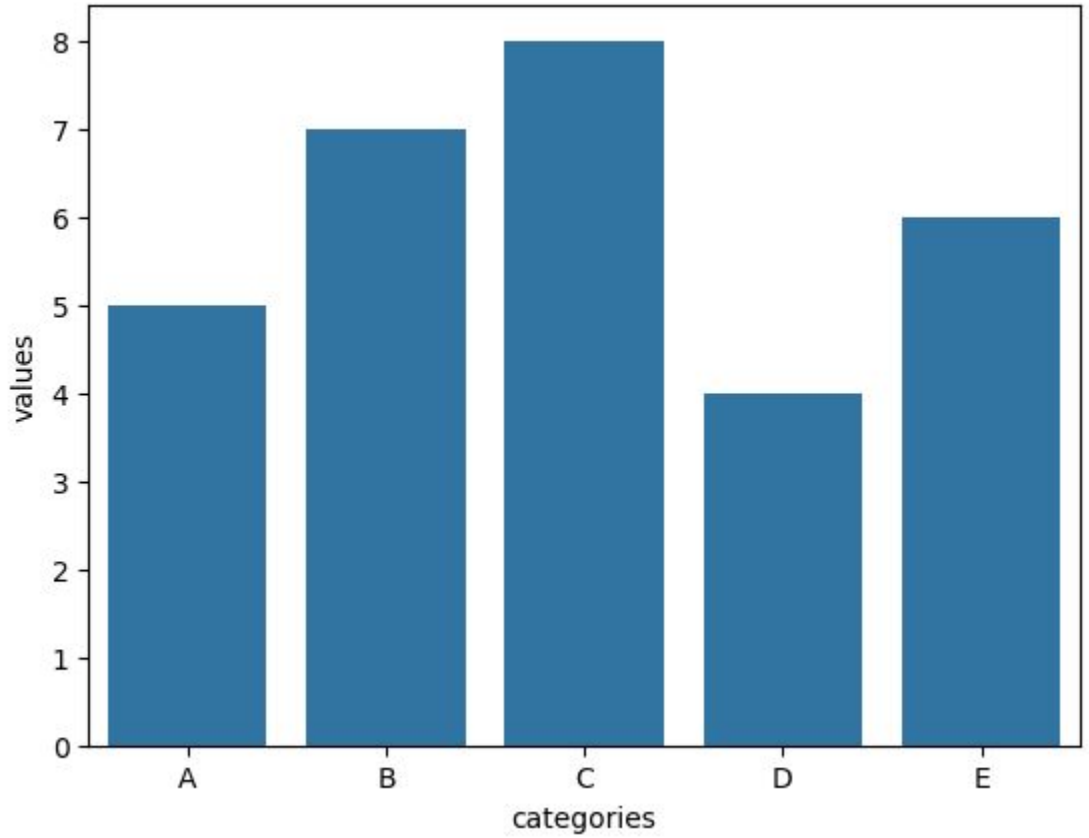
Scatter Plot

- **Description:** Scatter plots are used to represent individual data points, showing the relationship between two continuous variables.
- **Use Case:** Identifying correlations or patterns between two variables.



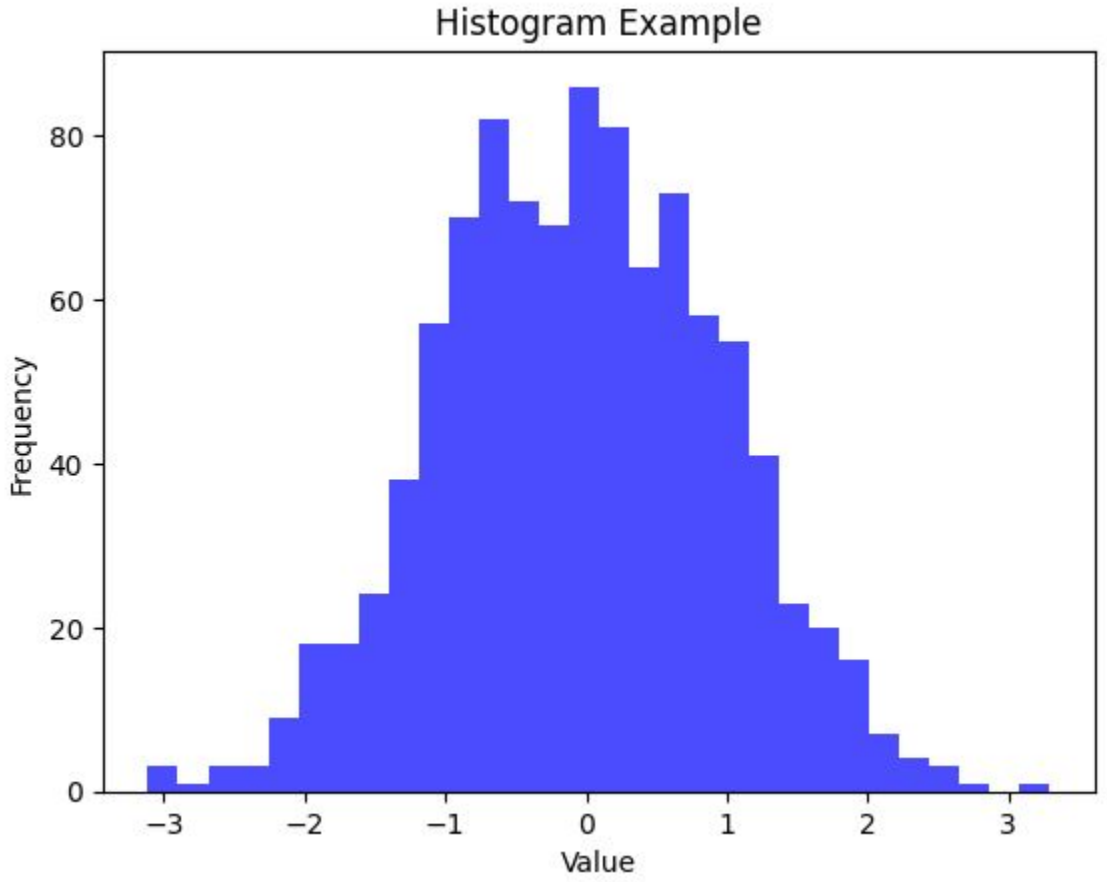
Bar Plot

- **Description:** Bar charts are used to compare different categories of data by representing them as rectangular bars with lengths proportional to the values they represent.
- **Use Case**: Comparing the frequency or magnitude of different categories.



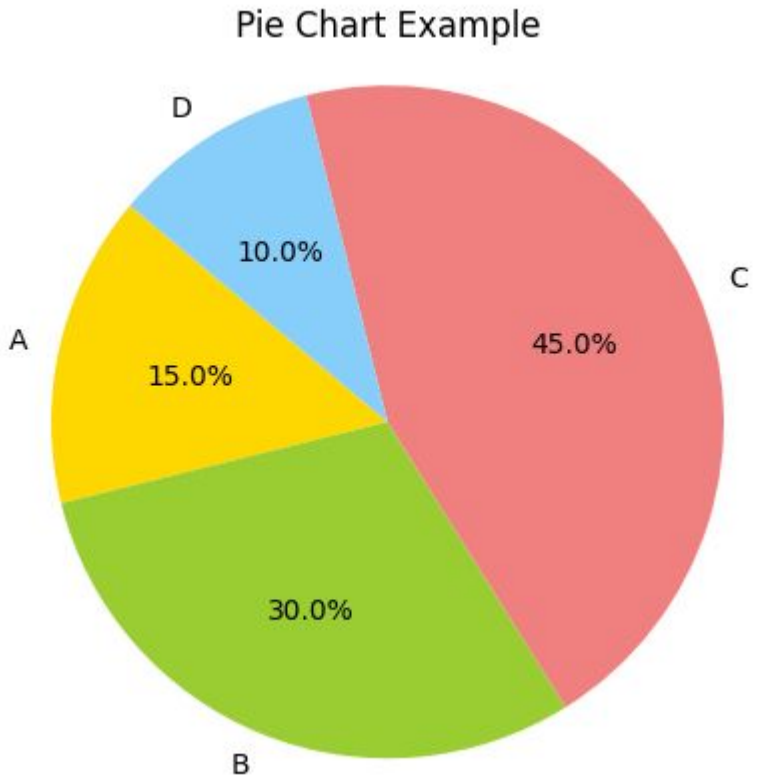
Histogram

- **Description:** Histograms show the distribution of a continuous variable by dividing the data into bins and displaying the frequency of data points in each bin.
- **Use Case:** Analyzing the distribution of a dataset.



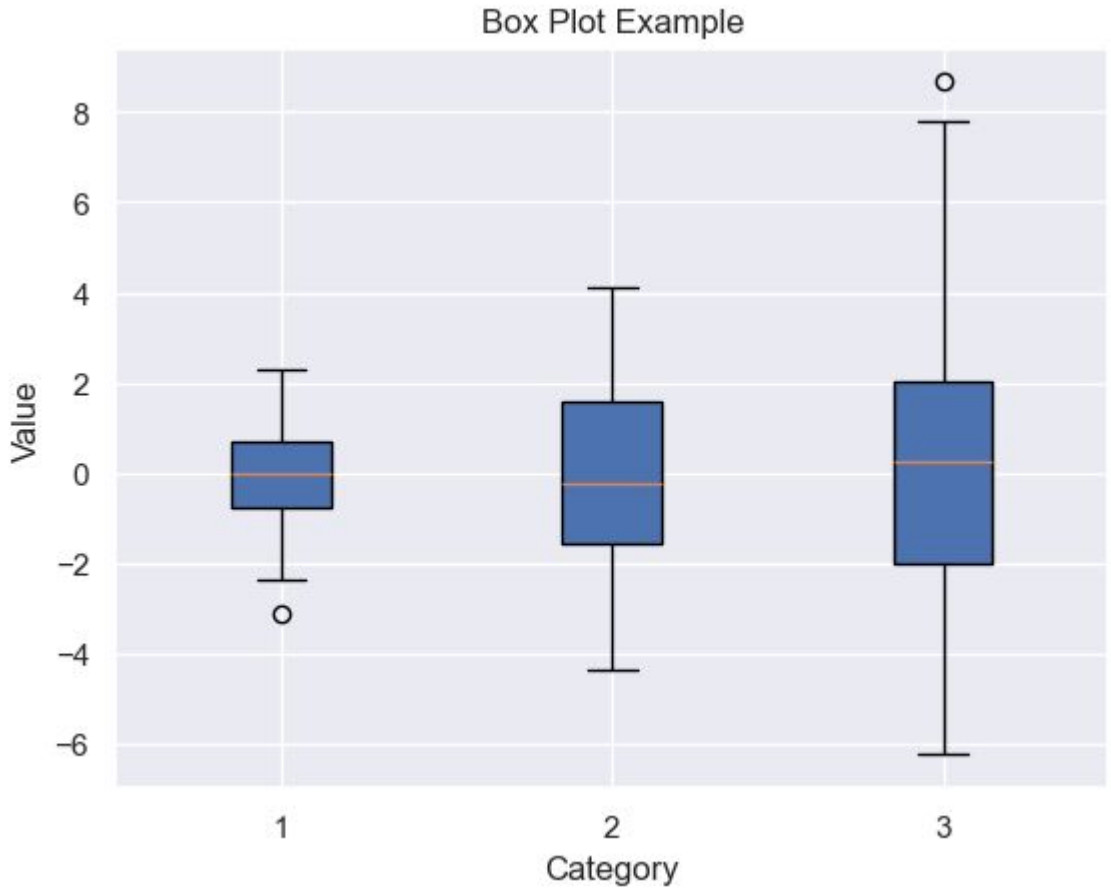
Pie Chart

- **Description:** Pie charts display data as segments of a circle, with each segment representing a proportion of the whole.
- **Use Case:** Representing the relative proportions of different categories in a dataset.



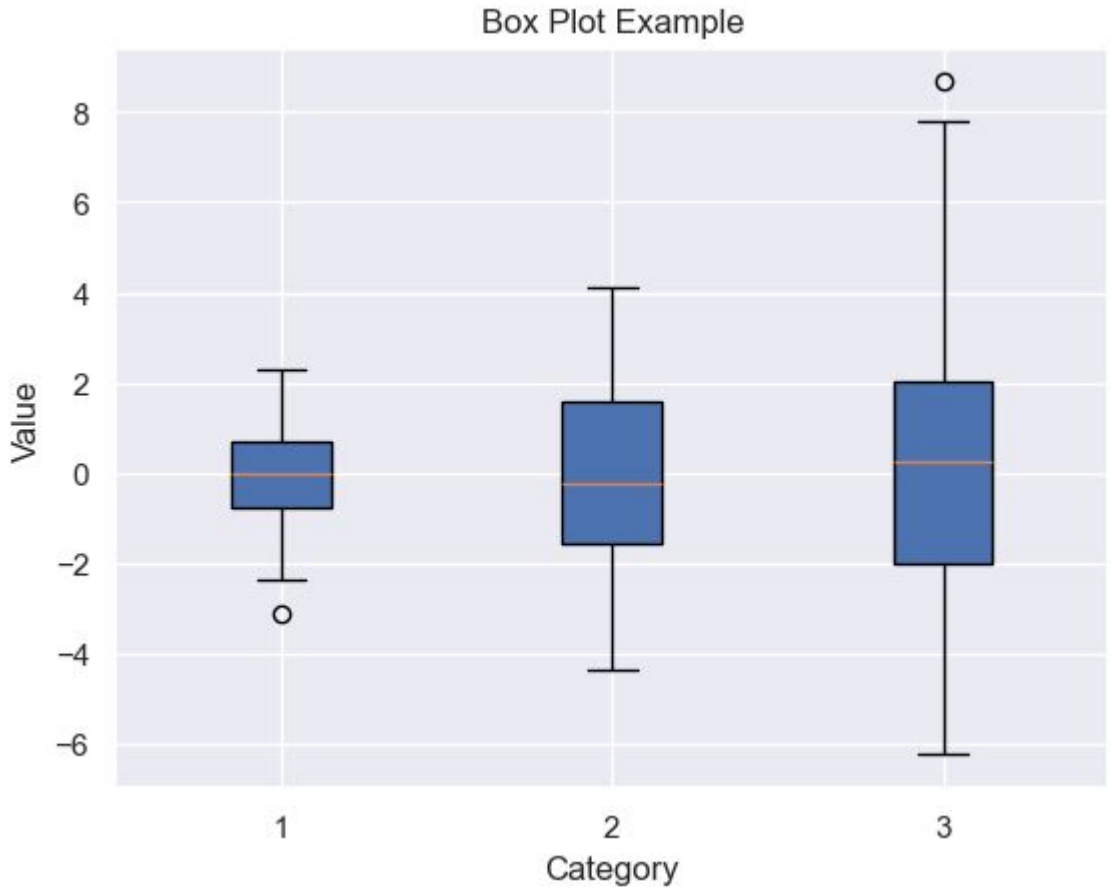
Box Plot

- **Description:** Box plots are used to display the distribution of a dataset based on five summary statistics: minimum, first quartile, median, third quartile, and maximum.
- **Use Case:** Comparing the distribution of multiple datasets or identifying outliers.



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CONCLUSION

This internship provided valuable hands-on experience in the domain of data science, particularly in data visualization. By completing the tasks outlined, I developed a strong understanding of Matplotlib and Seaborn, two powerful tools in the data scientist’s toolkit.

THANK YOU